

Model	Configuration		Evaluation	Performance		
	Strain features	Feature selection		Accuracy	ROC-AUC	F1-weighted
M1	Yes	Nested (within folds)	pair + strain held-out	0.50	0.52	0.48
M2	Yes	Nested (within folds)	pair-held-out	0.68	0.74	0.68
HALO	No	Nested (within folds)	pair-held-out	0.68	0.74	0.68
M3	No	Global (pre-CV)	pair-held-out	0.75	0.82	0.75
M4	No	Training-set only	Standard stratified split	0.76	0.84	0.76